Use *git checkout contact-form* to switch to the *contact-form* branch. To view the state of your project at a specific commit, use *git checkout <commit-hash>*.

Staging area: The staging area (index) allows you to prepare changes before committing them, letting you select specific changes for inclusion in a commit.

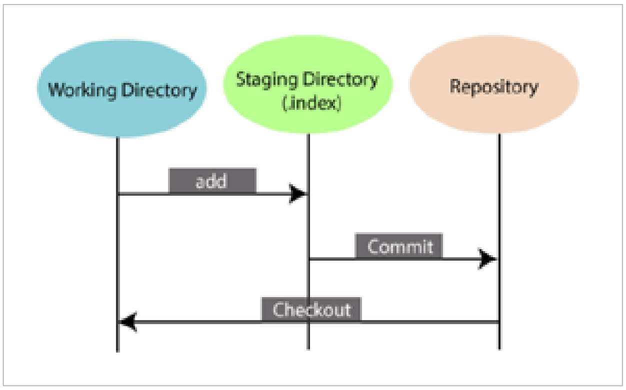

Figure 8: Staging area (Git)

After modifying multiple files, use *git add file1.txt* to stage changes to *file1.txt* and *git add file2.txt* for *file2.txt.* When you commit, only the staged files are included.

Diff: The *diff* command shows the differences between file versions, which is useful for reviewing changes before committing or merging.

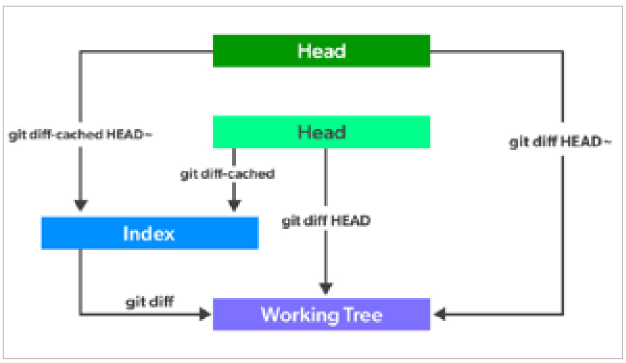

Figure 9: Diff command (Git)

To see changes between your working directory and the last commit, use *git diff*. To compare two commits, use *git diff <commit1> <commit2>*.

Basic overview of common Git workflows

Git offers flexibility in how teams can manage their code, but some workflows have become popular due to their effectiveness. Let's explore a few common ones.

- ***Feature branching***

Feature branching workflow involves creating separate branches for each new feature or bug fix. It keeps the main branch clean and stable.

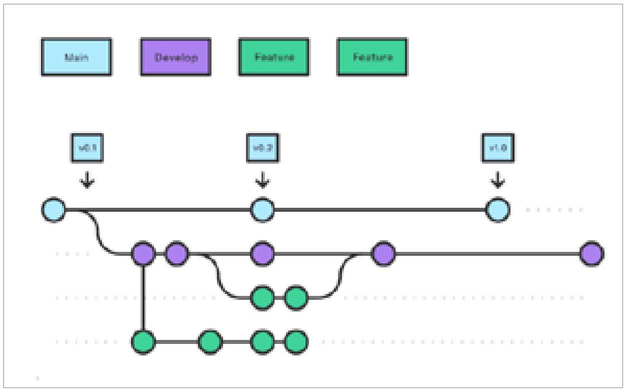

Figure 10: Feature branching

Process:

- Create a new branch from the *main* or *develop* branch for each new feature or fix.
- Work on the feature or fix in the new branch.
- Once complete, merge the feature branch back into the main branch, usually through a pull request or merge request.

Use case: Ideal for teams looking for a straightforward way to manage new features and fixes without disrupting the main codebase.

- **Git Flow**

Git Flow is a more structured branching model with specific roles for branches. It involves a set of branches with defined purposes and is particularly useful for managing releases and hotfixes.

Branches:

main (or master): The production-ready branch.

develop: The integration branch where features are merged.

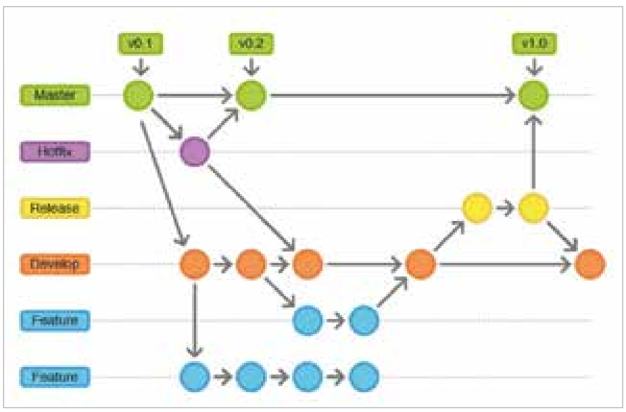

Figure 11: Git Flow